IT 119 - Data Scalability and Analytics

BSIT 3B

Renz Dominic Azul

Gabriel de la Cruz

Godwin Reigh Galvez

AI and the Environment: An Analysis of Public Awareness

Introduction

Artificial Intelligence (AI) has rapidly become an important part of everyday life, influencing education, communication, business, and other areas of society. However, the growing use of AI also brings environmental concerns because AI systems depend on data centers that consume large amounts of electricity and other resources. The International Energy Agency (IEA) reported that global data centers consumed approximately 415 terawatt-hours (TWh) of electricity in 2024, equivalent to about 1.5% of global electricity consumption, and this demand is projected to more than double by 2030, largely driven by AI. These developments highlight the importance of public awareness regarding the environmental costs associated with the increasing use of AI.

The issue is particularly relevant to the Philippines as the country expands its participation in the global AI and semiconductor industries. In April 2026, the Philippines joined Pax Silica, a United States-led initiative focused on strengthening supply chains for semiconductors and artificial intelligence. The initiative includes plans for a 4,000-acre industrial zone within the Luzon Economic Corridor that is intended to serve as an AI-native industrial acceleration hub. While such developments may create opportunities for technological advancement, investment, and employment, they also make it increasingly important for Filipinos to understand the possible environmental implications of expanding AI infrastructure and adoption. Public awareness can therefore contribute to more informed discussions about how technological development can be pursued while considering energy use, emissions, resource consumption, and environmental sustainability.

In this context, the study **“AI and the Environment: An Analysis of Public Awareness”** aims to determine the level of public awareness regarding the environmental impacts of AI. The study will examine respondents' knowledge and perceptions of issues related

to AI's energy consumption, environmental effects, and resource requirements. Through **descriptive, diagnostic, and exploratory analytics**, the researchers will describe the existing level of awareness, examine factors associated with differences in awareness, and explore patterns and relationships within the collected data. The findings may provide useful insights into how well the public understands the environmental implications of AI and contribute to discussions on responsible and sustainable AI adoption in the Philippines.

Objectives of the Study

To assess the level of community awareness regarding the environmental impact of Artificial Intelligence and determine its relationship with responsible AI usage.

Specific Objectives:

a. To determine the AI usage frequency of community residents.
b. To determine the level of awareness of community residents regarding the environmental impacts of Artificial Intelligence.
c. To assess the level of responsible AI usage among community residents.
d. To explore the environmental concerns and suggestions of community residents regarding the responsible use of Artificial Intelligence.

Significance of the Study

As AI is becoming increasingly integrated into everyday activities, its environmental implications may not be fully understood by ordinary users. Recent research on public awareness of AI's environmental impacts has found substantial differences in awareness among respondents. Cho et al. (2025) reported that only 21.9% of participants in Japan and 50.1% of participants in South Korea were aware of AI's environmental impacts, including its energy consumption, water use, and carbon emissions. The study suggests that although AI is increasingly being used in everyday activities, awareness of its environmental consequences may remain limited among users.

Furthermore, Jelonek and Pawełoszek (2026) also examined people's attitudes toward the environmental impact of AI and their willingness to adopt more sustainable AI-related behaviors. The researchers conducted a mixed-methods survey involving 224 respondents and

developed an Environmental Awareness Index that incorporated cognitive, evaluative, and behavioral aspects of environmental awareness. The study found that more than two-thirds of respondents expressed willingness to financially support greener AI solutions and demonstrates that environmental awareness can be examined through multiple dimensions rather than simply asking whether individuals know about AI's environmental impact.

As previously discussed, it is evident that the research about environmental awareness of AI is currently lacking. Therefore, this study will also examine whether greater environmental awareness is associated with more responsible AI usage, especially here in the Philippines, and contribute its findings to the community.

Research Design

This study will employ both quantitative, and mostly qualitative descriptive research design to assess the level of public awareness regarding the environmental impacts of Artificial Intelligence (AI). Data will be collected through a structured survey questionnaire consisting of questions about respondents' demographic characteristics, AI usage, knowledge, awareness, and perceptions of AI's environmental effects. The collected data will be cleaned, organized, and analyzed using *descriptive, diagnostic, and exploratory analytics. Descriptive analytics will be used to summarize the respondents' characteristics and overall level of awareness, diagnostic analytics will examine factors and relationships that may be associated with differences in awareness, and exploratory analytics will identify patterns or trends within the dataset. This research design will allow the researchers to obtain measurable evidence of public awareness and explore how different factors may relate to perceptions of AI and its environmental implications.

Data Gathering Procedure

The researchers will gather data through a structured questionnaire distributed to selected members of the target community. Before answering the questionnaire, respondents will be informed about the purpose of the study, the voluntary nature of their participation, and the intended use of the collected information. The questionnaire will contain demographic questions, AI usage questions, and primarily open-ended questions concerning respondents' knowledge, awareness, perceptions, and experiences regarding the environmental impacts of AI. The demographic section will collect information such as age, gender, and educational

attainment to allow the researchers to examine whether awareness and perceptions differ among different groups. The completed questionnaires will be collected, reviewed, and organized to ensure that the responses are suitable for analysis. Quantitative responses will be summarized using descriptive statistics, while open-ended responses will be reviewed and categorized according to recurring ideas, themes, and patterns. The resulting findings will then be examined to identify possible relationships and patterns between demographic characteristics, AI usage, and awareness of AI's environmental implications. Below are the following proposed questions:

Part I. Respondent Profile

1. What is your age?

- 18–24
- 25–34
- 35–44
- 45–54
- 55 and above

2. How often do you use Artificial Intelligence tools?

- Several times a day
- Once a day
- Several times a week
- Once a week
- Rarely

Part II. Environmental Awareness

5-point Likert scale:
Strongly Disagree – Disagree – Neutral – Agree – Strongly Agree

3. I am aware that using AI systems consumes significant amounts of electricity or energy.

4. I am aware that AI data centers can contribute to carbon emissions and other environmental impacts.

Part III. Responsible AI Usage

5. I consider the environmental impact of AI when deciding how I use AI tools.

6. I avoid unnecessary use of AI when the task can be completed without it.

Part IV. Qualitative Questions

7. What environmental impact of Artificial Intelligence concerns you the most, and why?

8. What do you think should be done to increase public awareness and encourage responsible use of AI?

References

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